

J.P Morgan ML Report: Part 1

SU SIQIANG

December 2025

a Are there any errors or unclear parts in the paper? If there are, please point them out and explain the unclear parts in the report.

- There is an notation confusion in Chapter 2.1. The original text is:

“A general and expressive framework for modeling such probabilities is McFadden’s universal logit model (McFadden et al., 1977; McFadden, 1984). For each $j \in \mathcal{S}$, define a utility function $u_j(T)$, which represents the utility of alternative j when the context is the subset $T \subseteq \mathcal{S} \setminus \{j\}$. Given a choice set S , the probability of choosing alternative $j \in S$ is defined as

$$P_j(S) = \frac{\exp(u_j(S))}{\sum_{k \in S} \exp(u_k(S))}.$$

”

The author introduce the notation T and $u_j(T)$. It is unclear that whether the utility in the equation should be $u_j(S)$ or $u_j(T)$? Should the utility of choice j depends on the choice j itself? So the introduction of subset T and $u_j(T)$ is misleading and unclear.

- There is a notation confusion in Chapter 3.1. The original text is:

“that will be discussed in the next subsection, we treat the choice set S as an ordered tuple of indices $(1, \dots, |S|)$, and represent the corresponding input feature matrix as

$$X_S := [x_1, \dots, x_{|S|}, O_{J-|S|}] \in \mathbb{R}^{d \times J},$$

where $O_{J-|S|}$ is a $d \times (J - |S|)$ zero matrix to pad the feature matrix to a fixed size $d \times J$. ”

In the first paragraph of Chapter 3, author already introduced the feature vector as $x_j \in \mathbb{R}^{d_x}$. The number of the features is d_x . For notation consistency, the dimension of the matrix X_S should be $d_x \times J$.

- The notation in Equation 3 is unclear. The original equation is:

“

$$v_j(X_{T \cup \{j\}}) := \sum_{R \subset T} (-1)^{|T|-|R|} u_j(x_j, X_R).$$

”

It is unclear that why the utility function u_j now takes two arguments x_j and X_R . It should be $u_j(X_{R \cup \{j\}})$ instead.

- There is a notation confusion in Chapter 3.2. The original text is:

“Let $z^0 := [\chi(x_j)]_{j \in S}$, where $\chi : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_0}$ is a shared nonlinear embedding function applied to each alternative. The use of a shared transformation ensures permutation equivariance. We define a context summary vector by linearly aggregating the embedded representations of all alternatives:

$$\bar{Z}^1 := \frac{1}{S} \sum_{k \in S} W^1 z_k^0 \in \mathbb{R}^H,$$

where $W^1 \in \mathbb{R}^{H \times d}$ is a shared linear projection across ...”

The matrix W_1 should be in dimension $H \times d_0$, not $H \times d$. Also, the multiplying factor should be $\frac{1}{|S|}$ instead of $\frac{1}{S}$, since S is a set.

- The multiplying factor before summation in Equation 4 and 6 are wrong, it should be $\frac{1}{|S|}$ instead of $\frac{1}{S}$.
- There is a mistake in Equation 7. The original equation is the following:

“

$$z_j^l = z_j^{l-1} + \frac{1}{H} \sum_{h=1}^H \Phi_h^l(\bar{Z}_h^l, \phi_h^l(z_j^0))$$

where Φ_h^l is a polynomial function on $\mathbb{R} \times \mathbb{R}$.”

The function Φ_h^l takes two arguments $\bar{Z}_h^l \in \mathbb{R}$ and $\phi_h^l(z_j^0) \in \mathbb{R}^{d_0}$, so it should be a function with the domain defined on $\mathbb{R} \times \mathbb{R}^{d_0}$ instead of $\mathbb{R} \times \mathbb{R}$.

b Do you get the same results as their synthetic data test?

I implemented the synthetic data set experiment and the LPMC real data experiment, using the hyper-parameter settings stated in the paper Appendix C.2.2.

For synthetic data experiment, due to the computation challenge and time limit, I select a subset of the training data to reduce the sample size to accelerate the computation. The training RMSE is 0.0467, which is quite close to the results in the paper.

It is important to note that due to computational constraints and time limits, our training was performed on a subset of the full synthetic training data to accelerate the computation. While this approach allowed for timely validation of the model's functionality, it might lead to minor deviations from the paper's exact figures, which used the complete dataset.

For the LPMC real data experiment, our implementation yielded a training NLL of 0.7941 and a testing NLL of 0.7918. These results are inferior to those reported in the paper (training: 0.6427, testing: 0.6407), indicating a significant discrepancy in performance.

Those discrepancy is a common challenge in replicating complex deep learning models and can stem from a multitude of subtle factors. Potential contributors to this difference include, but are not limited to:

- **Data Preprocessing:**

- **Categorical Encoding:** Subtle differences in how categorical features (e.g., choice attributes, user demographics) are one-hot encoded or embedded.
- **Dataset Version:** While using the LPMC dataset, there might be slight variations in the version or specific cleaning steps applied that were not fully detailed.

- **Model Implementation Details:**

- **Activation Functions:** Minor variations in the exact implementation of activation functions or their placement within the network.
- **Weight Initialization:** The specific random seed used for initializing model weights, or the default initialization schemes of different deep learning frameworks, can lead to varying training paths.

The implementation code can be accessed through the Github page <https://github.com/siqiang-su/J.P-Morgan-ML-Project>.

c What additional tests could you come up with to verify the results?

To thoroughly verify the claims and efficacy of the Deep Context-Dependent Choice Model, the following additional tests would be beneficial:

1. **Quantifying Benefits of Explicit Interaction Order Control:** The authors claim that their method offers exact control over interaction order, unlike other deep learning models that often entangle interactions. To quantify the practical benefits of this control, comparative experiments are needed. For a given interaction order L , DeepHalo’s performance should be compared against other deep choice models whose results are truncated up to the interaction order L . This would reveal if explicit control leads to superior performance.
2. **Test RMSE for Synthetic Data with Higher-Order Effects:** The synthetic data experiment (Section 5.1, Figures 2 & 3) only reports training RMSE. To fully verify the efficacy and generalization capability of the proposed method, especially its ability to capture higher-order effects without overfitting, the testing RMSE on unseen synthetic data is crucial. This would confirm that the model learns generalizable context effects rather than merely memorizing the training examples.
3. **Sensitivity Analysis and Feature Importance for Feature-Based Datasets:** For feature-based datasets, a systematic perturbation analysis of specific features of alternatives is needed. By making controlled changes to individual feature values of specific items in a choice set, one can observe the corresponding impact on predicted choice probabilities and the identified relative context effects.
4. **Computational Scalability Analysis:** While the paper discusses efficiency, a more explicit comparison of computational resources against state-of-the-art baselines would be valuable. This analysis should cover scenarios with increasing choice set sizes, feature dimensions, and desired interaction orders, providing practical guidance for deployment in real-world large-scale applications.

d Comparing to the reproducing kernel Hilbert Space Choice model of Yang(25), what are the pros and cons of Zhang(25)?

Introduction to reproducing kernel Hilbert Space Choice model

We begin by defining a vector-valued RKHS on the space of sets. Let $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ denote a $\mathbb{R}^J$ -valued Hilbert space equipped with the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and the induced norm $\| \cdot \|_{\mathcal{H}}$. A function $K : \mathcal{S} \times \mathcal{S} \rightarrow \mathbb{R}^{J \times J}$ is defined as a kernel if it satisfies the properties of symmetry and positive definiteness. The kernel function $K(x_i, x_j)$, which accepts two inputs, serves as a measure of similarity between items

x_i and x_j . Consequently, the kernel provides a principled approach for incorporating contextual effects in choice modeling.

The Hilbert space $\mathcal{H}$ constitutes an RKHS if it satisfies the reproducing property, which establishes that for every choice set $S \subset \mathcal{S}$ and every utility function $u \in \mathcal{H}$, the evaluation of utility u at S can be expressed as an inner product in $\mathcal{H}$:

$$u_j(S) = \langle u, K(\cdot, S)e_j \rangle_{\mathcal{H}}, \quad j \in S, \quad (1)$$

where e_j denotes the j -th standard basis vector with unity in the j -th position and zeros elsewhere.

Rather than modeling the utility function u through parametric specifications such as linear models, the RKHS choice model employs a non-parametric, model-free approach. The space of candidate utility functions encompasses all functions within the Hilbert space $\mathcal{H}$. Accordingly, the objective function for the RKHS choice model is formulated as:

$$\min_{u \in \mathcal{H}_{\hat{K}}} \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, p^i(u)) + \lambda \|u\|_{\mathcal{H}_{\hat{K}}}^2. \quad (2)$$

This optimization problem seeks to identify an optimal function within an infinite-dimensional space, a task that is generally intractable through conventional methods. However, a fundamental property of RKHS ensures that the optimal function u^* can always be expressed as a finite weighted sum of kernel functions centered at the observed data points. Specifically, the utility function admits the representation:

$$u(\cdot) = \sum_{i=1}^N \bar{K}\left(\cdot, \left(\hat{S}_i, \hat{X}_{\hat{S}_i}\right)\right) \alpha_i. \quad (3)$$

This representer theorem enables the reformulation of the objective function as a finite-dimensional optimization problem:

$$\min_{\alpha = \{\alpha_i\}_{i=1}^N \subset \mathbb{R}^J} \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, p^i(u(\cdot; \alpha))) + \lambda \sum_{i=1}^N \sum_{j=1}^N \alpha_i^\top \bar{K}\left(\left(\hat{S}_i, \hat{X}_{\hat{S}_i}\right), \left(\hat{S}_j, \hat{X}_{\hat{S}_j}\right)\right) \alpha_j, \quad (4)$$

which becomes computationally tractable through standard optimization techniques.

Given this theoretical framework, a critical consideration concerns the selection of the kernel function K . The literature presents three principal approaches:

1. **Pre-defined kernel functions:** This approach employs established kernel functions such as polynomial, Gaussian, or Matérn kernels. While conceptually straightforward, this method necessitates the *a priori* computation and storage of kernel function values. For large choice sets, substantial computational

and memory resources are required to calculate and maintain these kernel evaluations.

2. **Random feature approximation:** This methodology approximates the kernel through random feature methods, reducing computational and memory complexity from $O(N^2J^2)$ to $O(N_wNJ)$, where N represents the sample size, J denotes the choice set size, and N_w corresponds to the number of discretization points for the random distribution. However, this computational efficiency may come at the cost of reduced approximation accuracy.
3. **Neural Tangent Kernel (NTK):** This approach parametrizes the kernel function through a neural network architecture incorporating self-attention mechanisms, enabling adaptive learning of the kernel structure from the data. This methodology provides substantial flexibility in kernel specification, thereby enhancing the model’s expressive capacity.

Finally, the authors establish theoretical guarantees through convergence theorems that quantify the convergence rate of stochastic gradient descent optimization when applied to NTK training, and provide generalization error bounds for the RKHS choice model framework.

Pros and cons of Zhang(25)

We first discuss the pros of Zhang(25) comparing to Yang(25):

1. **Explicit Control over Interaction Order:** A key advantage of Zhang (2025) is the explicit control over the interaction order among alternatives. The model expresses the utility as a sum of interactions of increasing order:

$$u_j(X_S) = \sum_{p=0}^{|S|-1} \sum_{T \subset S \setminus \{j\}, |T|=p} v_j(X_{T \cup \{j\}}).$$

This decomposition allows practitioners to control the maximum interaction order by truncating the sum at a desired level. In contrast, the RKHS choice model captures context effects via a kernel function K , which inherently entangles all possible interactions. For example, a polynomial kernel aggregates all interactions up to a specified order p , but does not allow for explicit truncation or separation of interaction orders.

2. **Enhanced Interpretability:** In many practical applications, researchers are primarily interested in lower-order interactions. Zhang (2025) enables explicit modeling and interpretation of these interactions, which can be particularly meaningful for empirical analysis and policy interpretation.

Next we discuss the cons of Zhang(25) comparing to Yang(25):

1. Increased Computational Complexity for High-Order Interactions:

Capturing higher-order interactions in Zhang’s framework requires a recursive neural network architecture, which can significantly increase the depth and complexity of the model, leading to greater computational burden during training. In contrast, the RKHS approach is computationally more efficient, as evaluating kernel functions (such as polynomial kernels for up to p -th order interactions) is generally simpler and less resource-intensive than training deep neural networks.

2. Limited Theoretical Guarantees: Zhang’s model, being based on deep neural networks, involves highly non-convex objective functions, making it challenging to establish theoretical results regarding optimization convergence and generalization error bounds. On the other hand, the RKHS choice model benefits from the mathematical properties of RKHS, reducing the learning problem to a convex optimization. This allows for provable convergence rates for stochastic gradient methods (e.g., the expected loss approaches the optimal loss at a rate of $O(1/\sqrt{K})$ as the number of iterations K increases), and generalization error bounds can be controlled by the RKHS norm of the utility functions, typically of order $O(\sqrt{J/N})$.

e Do you think these models are suitable for the task of demand estimation of credit card offers? What are the potential problems or challenges?

Overview: demand estimation of credit card offers

A critical step in forecasting the demand for credit card promotional offers is the selection of an appropriate discrete choice model. Recent methodological advances, such as the deep context-dependent choice model (Zhang, 2025) and the RKHS choice model (Yang, 2025), offer sophisticated frameworks for capturing complex consumer decision rules. This section evaluates the suitability of these models for credit card offer demand estimation, outlining their theoretical alignment with key industry challenges and the practical considerations for their application.

The modeling task begins by defining the universal set of potential offers, $\mathcal{S}$, from which subsets are presented to customers. Each offer alternative j is characterized by a feature vector $\mathbf{x}_j$, encompassing attributes such as redemption value (cashback or discount), minimum spend threshold, complexity of the redemption process, partner vendor or product category, and offer duration or expiration.

Concurrently, customer-specific characteristics, denoted by a vector $\mathbf{z}_i$, are crucial

for personalized prediction. Key variables include credit score and risk tier, income bracket, tenure with the issuing bank, and historical spending patterns.

The objective is to estimate the probability P_{ij} that customer i chooses offer j from a presented choice set $S \subset \mathcal{S}$, conditional on both $\mathbf{x}_j$ and $\mathbf{z}_i$.

Standard discrete choice models (e.g., Multinomial Logit) often rely on the assumption of Independence from Irrelevant Alternatives (IIA), which posits that the relative preference between two offers is unaffected by the presence or absence of other options. This assumption is frequently violated in real-world credit card offer portfolios due to strong Halo effects.

For example, the Decoy Effect describes a scenario where a strategically designed, dominated offer alters the attractiveness of others. For instance, introducing a decoy offer (“Spend \$80, get \$8 back”) can increase the perceived value and selection probability of a target offer (“Spend \$100, get \$12 back”) relative to a competitor (“Spend \$50, get \$6 back”). Conversely, the Compromise Effect occurs when an offer gains significant choice share by being presented as a middle option between two extremes. A mid-tier offer (“Spend \$75, get \$9 back”) will often see increased demand when flanked by a low-value (“Spend \$25, get \$2 back”) and a high-value, high-threshold (“Spend \$150, get \$15 back”) alternative.

The deep context-dependent model (Zhang, 2025) and the RKHS choice model (Yang, 2025) are theoretically well-suited to address these challenges. The former uses deep neural networks to learn complex, non-linear interactions between offer features and the choice context directly from data. The latter employs Reproducing Kernel Hilbert Space (RKHS) techniques to model subtle similarities and dependencies between alternatives in a high-dimensional feature space, naturally accounting for context effects like compromise and similarity.

Potential problems and challenges

1. Credit card offers are highly dynamic, with new offers constantly appearing and old ones expiring. The model would need to be frequently retrained or adapted to handle this changing landscape, potentially requiring online learning capabilities or robust transfer learning.
2. For deep-learning based models, it requires a lot of amount training data. However, for new offers or new merchants, where historical data is scarce, the model might struggle to accurately predict demand without sufficient training data.
3. While the model can incorporate item features, but it does not mention explicitly how individual features are modeled. In reality, the individual features like age, income, sex, preference of merchat will affect their choice largely.

4. For credit card offers, customers might choose none of the offers, or might also accept multiple offers. The Deep Context-Dependent Choice Model assumes that customer always make exact one choice among the available choice set.

f What are other models, traditional non-neural network based, or ML models that you think are more suitable or at least worth trying? Why?

Credit card offer demand estimation can be framed as a choice prediction problem, which shares significant characteristics with a multi-class classification task. Given J alternatives (offers), along with their item-specific features and individual customer features, the goal is to predict which alternative a customer is most likely to choose, or if they choose none (the 'no-choice' option). Crucially, for demand estimation, the output needs to be well-calibrated probabilities of choosing each option, rather than just a single predicted class label. Consequently, various non-neural network-based Machine Learning classification methods can be deployed, often by adapting their input structure to reflect choice sets. These include Gradient Boosting Machines (GBMs) like XGBoost.

Below, we discuss both traditional non-neural network based choice models and general-purpose machine learning models that can be adapted for this task.

Non-Neural Network Based Choice Models

- **Context Logit / Halo Model** (Maragheh et al., 2018; Seshadri et al., 2019; Yousefi Maragheh et al., 2020):
 - * **Why worth trying:** These models are specifically designed to explicitly incorporate first-order pairwise context effects. They offer a good balance between interpretability and computational efficiency for simpler interactions. For scenarios where pairwise interactions are dominant and sufficient for insights, these models can be very effective.
 - * **Pros:** These models offer higher computational efficiency compared to neural-network based choice models. In many real applications, understanding first-order pairwise context effects provides sufficient and actionable insights, as higher-order context effects can be more challenging to interpret or less economically meaningful to pursue. They also retain a degree of interpretability akin to traditional logit models for direct effects.
 - * **Cons:** These models typically restrict interactions to first-order pairwise terms, potentially missing crucial higher-order interactions that could be

important for complex offer combinations. They also rely on pre-defined functional forms for utility, which can limit their flexibility in capturing complex, non-linear relationships within rich feature sets compared to more data-driven ML approaches.

Machine Learning Models (Adapted for Choice Problems)

– Gradient Boosting Machines (GBMs):

- * **Why worth trying:** Gradient Boosting Machines (GBMs), exemplified by XGBoost, are powerful ensemble learning methods. They build trees sequentially, with each new tree correcting the errors of the previous ones, allowing them to learn complex patterns and achieve high predictive accuracy. For credit card offer demand estimation, the problem is typically formulated as an item-centric binary classification or ranking task. Here, the model predicts the likelihood of choosing each individual offer within a choice set, taking into account customer features, offer features, and carefully engineered contextual features derived from other offers in the set.
- * **Pros:** GBMs like XGBoost are known for their high predictive accuracy and ability to capture complex, non-linear interactions between features. They can seamlessly handle both numerical and categorical features and provide valuable insights through feature importance scores. They are often a strong baseline and can achieve state-of-the-art performance on tabular data.
- * **Cons:** Despite their predictive power, GBMs are generally considered “black-box” models. Unlike traditional choice models (e.g., MNL) that provide interpretable coefficients reflecting utility contributions, GBMs lack this direct interpretability, making it harder to understand the underlying choice drivers. Furthermore, GBMs are not inherently permutation invariant regarding the order of items in a choice set.